Biology Bridge Course Textbook

High School Biology Preparation - Chapters, Diagrams, Memory Tricks, Virtual Labs, Exam Q&A;, Mock Exams, and Challenge Problems

How to use this textbook

This guide is designed as a bridge course for a student preparing for high school biology. It starts with the building blocks of life and gradually connects cells, DNA, ecosystems, evolution, energy, and human body systems. Each chapter includes explanations, visual summaries, memory tricks, lab-style reasoning, and exam-style questions with answers.

Recommended study method: read the chapter once, cover the answer section, answer the questions like an exam, then check your reasoning. For the diagrams, practice redrawing them from memory. That is one of the fastest ways to move from memorizing biology to understanding biology.

Course Map

Chapter	Main Focus	Why it matters
1	Cells	All organisms are made of cells; organelles explain life functions.
2	DNA and Heredity	DNA stores instructions and explains inheritance.
3	Ecosystems	Energy flow and interactions explain environmental balance.
4	Evolution and Natural Selection	Explains how populations change over time.
5	Energy in Organisms	Photosynthesis and respiration power life.
6	Human Body Systems	Body systems work together to maintain homeostasis.
Exam Pack	Mock exams and harder questions	Builds test readiness and reasoning skills.

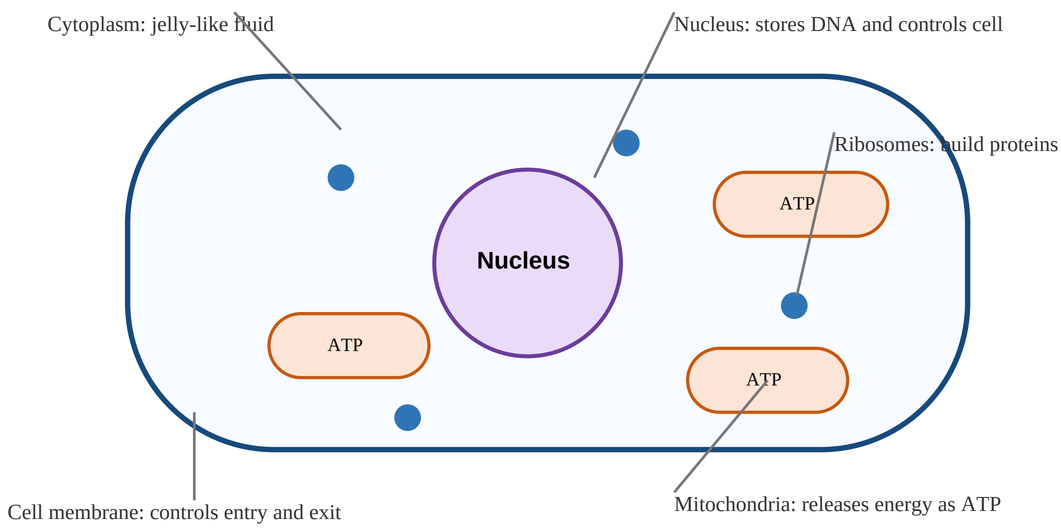
Chapter 1: Cells - The Basic Unit of Life

Core idea: All living things are made of cells, and cells are the smallest units that can perform life processes.

Learning Goals

- Explain the three parts of cell theory.
- Compare prokaryotic and eukaryotic cells.
- Identify major organelles and connect each organelle to its function.
- Use microscope-style clues to classify a cell.

Animal Cell - Major Organelles



Step-by-Step Explanation

1. Cell Theory

Cell theory has three major statements: all living things are made of cells; cells are the basic unit of structure and function in living things; and all cells come from pre-existing cells. This idea is important because it gives biology a starting point: to understand life, first understand cells.

2. Prokaryotic vs. Eukaryotic Cells

Prokaryotic cells are usually smaller and simpler. They do not have a nucleus, so their DNA is located in the cytoplasm. Bacteria are prokaryotes. Eukaryotic cells have a nucleus and membrane-bound organelles. Plants, animals, fungi, and protists are eukaryotes.

3. Organelles

The nucleus stores DNA and helps control cell activities. Mitochondria release energy from food and produce ATP. Ribosomes build proteins. The cell membrane controls what enters and leaves. Cytoplasm is the jelly-like interior where many reactions happen. Plant cells also have a cell wall, chloroplasts, and a large central vacuole.

Key Vocabulary

Term	Meaning
Cell	Smallest unit of life that can carry out life functions.
Organelle	A specialized structure inside a cell.
Prokaryote	Cell without a nucleus, such as bacteria.
Eukaryote	Cell with a nucleus, such as plant and animal cells.
ATP	Usable energy molecule for cells.

Visual Summary and Memory Tricks

Cell = City: nucleus = mayor, mitochondria = power plant, ribosomes = factories, cell membrane = city gate, cytoplasm = city space.

Prokaryote clue: no nucleus. Eukaryote clue: nucleus present. Plant clue: cell wall + chloroplasts + large vacuole.

Virtual Lab / Investigation Practice

Virtual microscope scenario: You observe a rectangular cell with a thick outer boundary, green structures, and a large vacuole. This is likely a plant cell because the thick boundary is a cell wall, the green structures are chloroplasts, and the large vacuole stores water.

Data analysis idea: If a cell has many mitochondria, it probably needs a lot of energy. Muscle cells often have many mitochondria because movement requires ATP.

Exam-Style Question and Answer Session

Q1. What is the function of the mitochondrion?

Answer: It releases energy from food molecules and produces ATP for cell work.

Q2. What makes a cell eukaryotic?

Answer: A eukaryotic cell has a nucleus and membrane-bound organelles.

Q3. Why is the cell membrane important?

Answer: It maintains the cell boundary and controls what enters and leaves the cell.

Q4. A cell has no nucleus. What type is it most likely?

Answer: It is most likely prokaryotic.

Q5. Why do plant cells need chloroplasts?

Answer: Chloroplasts allow plant cells to perform photosynthesis and make glucose.

Harder Challenge Question

Question: A scientist finds a cell with DNA, ribosomes, cytoplasm, and a membrane but no nucleus. Is it alive, and how would you classify it?

Answer: Yes, it can be alive. It is likely a prokaryotic cell because it has cellular structures but lacks a nucleus.

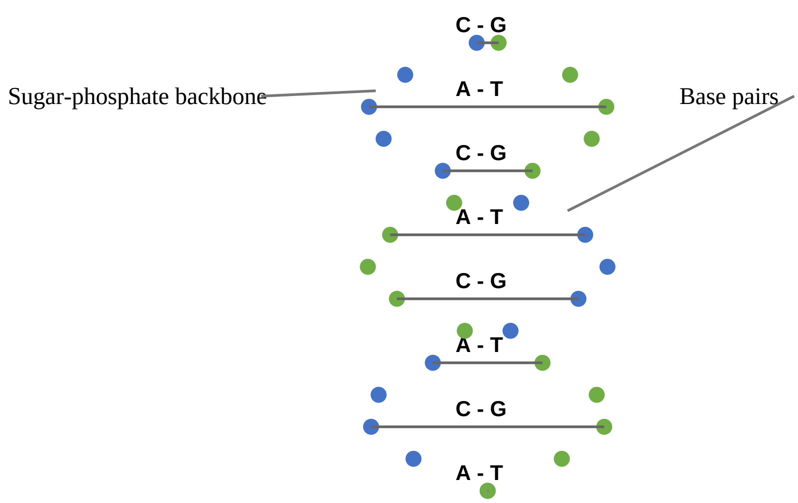
Chapter 2: DNA and Heredity

Core idea: DNA stores genetic instructions, and genes influence traits passed from parents to offspring.

Learning Goals

- Describe the structure of DNA.
- Apply base-pairing rules.
- Explain genes, chromosomes, alleles, genotype, and phenotype.
- Use Punnett squares to predict simple inheritance patterns.

DNA Double Helix and Base Pairing



Step-by-Step Explanation

1. DNA Structure

DNA is shaped like a double helix, similar to a twisted ladder. The sides are sugar-phosphate backbones, and the rungs are nitrogen base pairs. The pairing rule is A with T and C with G.

2. Genes and Traits

A gene is a section of DNA that helps code for a protein or functional molecule. Proteins influence traits. A trait such as eye color or blood type is connected to genes, but many traits are influenced by multiple genes and the environment.

3. Inheritance

Parents pass alleles to offspring. A Punnett square predicts possible allele combinations. For example, if B is dominant for brown eyes and b is recessive, a Bb parent can pass either B or b. Dominant traits show when at least one dominant allele is present. Recessive traits usually show only when both alleles are recessive.

Key Vocabulary

Term	Meaning

DNA	Molecule that stores genetic instructions.
Gene	Segment of DNA that influences a trait.
Chromosome	Long packaged DNA structure.
Allele	Different form of a gene.
Genotype	Allele combination.
Phenotype	Observable trait.
Mutation	Change in DNA sequence.

Visual Summary and Memory Tricks

DNA = instruction book. Gene = chapter. Chromosome = volume of the book. Trait = result of instructions being used.

Base-pair cue: Apple Tree means A pairs with T. Car Garage means C pairs with G.

Genotype is the letters; phenotype is what you see.

Virtual Lab / Investigation Practice

Virtual lab scenario: A DNA sequence reads A T C G. The complementary strand is T A G C. If one base changes, a mutation has occurred. The mutation may have no effect, a harmful effect, or rarely a helpful effect depending on how it changes the protein.

Punnett practice: Cross Tt x Tt. Possible genotypes are TT, Tt, Tt, tt. If T is dominant, the phenotype ratio is 3 dominant to 1 recessive.

Exam-Style Question and Answer Session

Q1. What does DNA do?

Answer: DNA stores genetic instructions used to build and regulate living organisms.

Q2. What are the base-pairing rules?

Answer: A pairs with T, and C pairs with G.

Q3. What is the difference between genotype and phenotype?

Answer: Genotype is the allele combination; phenotype is the observable trait.

Q4. What is a mutation?

Answer: A mutation is a change in the DNA sequence.

Q5. If a dominant allele is represented by A, which genotypes show the dominant phenotype?

Answer: AA and Aa show the dominant phenotype.

Harder Challenge Question

Question: Two heterozygous parents, Aa and Aa, have a child. What is the probability that the child has genotype aa?

Answer: The Punnett square outcomes are AA, Aa, Aa, and aa. The probability of aa is 1 out of 4, or 25%.

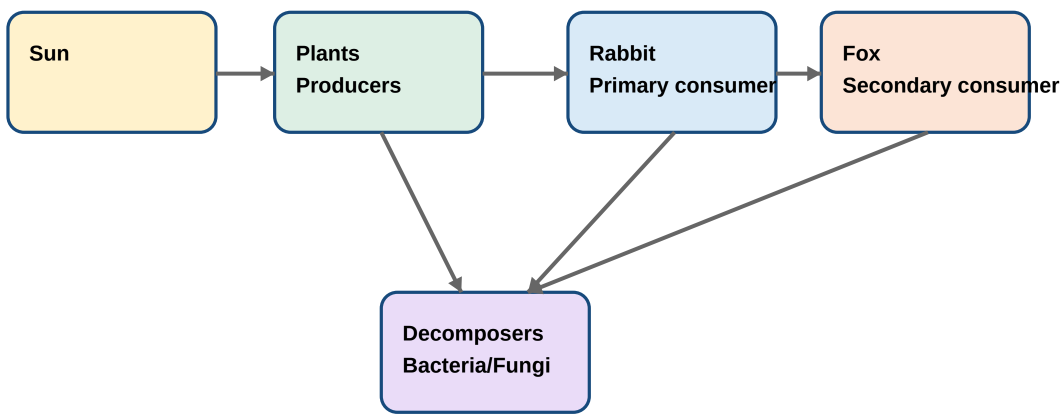
Chapter 3: Ecosystems - Interdependence of Life

Core idea: An ecosystem includes living and nonliving parts interacting through energy flow and matter cycling.

Learning Goals

- Distinguish biotic and abiotic factors.
- Trace energy through food chains and food webs.
- Explain producers, consumers, and decomposers.
- Predict how changes affect populations in an ecosystem.

Ecosystem Energy Flow



Key idea: Energy mostly flows one way. Matter and nutrients recycle through decomposers.

Step-by-Step Explanation

1. Components of an Ecosystem

Biotic factors include plants, animals, fungi, bacteria, and dead organisms. Abiotic factors include sunlight, water, oxygen, minerals, temperature, and soil. Both matter. A plant depends on sunlight and water, while animals depend on plants or other animals.

2. Energy Flow

Energy enters most ecosystems from the Sun. Producers capture solar energy and store it in glucose. Consumers get energy by eating producers or other consumers. Energy decreases at higher trophic levels because organisms use energy for movement, heat, growth, and survival.

3. Food Webs and Stability

A food chain shows one path of energy transfer. A food web shows many connected feeding relationships. Food webs are more realistic because organisms often eat more than one type of food. If one species changes, several other populations may be affected.

Key Vocabulary

Term	Meaning
Ecosystem	All living and nonliving things interacting in an area.
Biotic	Living or once-living factor.
Abiotic	Nonliving factor such as water, sunlight, temperature, or soil.
Producer	Organism that makes its own food, usually by photosynthesis.
Consumer	Organism that eats other organisms.
Decomposer	Organism that breaks down dead material and recycles nutrients.

Visual Summary and Memory Tricks

Sun starts everything: Sun -> Producers -> Consumers -> Decomposers.

Food chain = one road. Food web = many connected roads.

Decomposers are nature recycle teams.

Virtual Lab / Investigation Practice

Virtual ecosystem simulation: If rabbits disappear, plants may increase because fewer rabbits eat them, while foxes may decrease because they lose a food source.

Data interpretation: If a graph shows algae increasing after fertilizer runoff, the ecosystem may experience algal bloom, oxygen depletion, and fish death.

Exam-Style Question and Answer Session

Q1. What is an ecosystem?

Answer: An ecosystem is the interaction of living and nonliving things in an area.

Q2. What is the difference between biotic and abiotic factors?

Answer: Biotic factors are living or once-living; abiotic factors are nonliving.

Q3. Why are producers important?

Answer: They convert sunlight into stored chemical energy that supports food webs.

Q4. What do decomposers do?

Answer: They break down dead organisms and recycle nutrients.

Q5. What might happen if a top predator disappears?

Answer: Prey populations may increase, which can affect plants and other organisms.

Harder Challenge Question

Question: A disease kills most plants in a grassland. Explain two effects on the food web.

Answer: Herbivores may decrease because they lose food. Predators may also decrease later because they lose prey. Oxygen and nutrient cycling may also be affected.

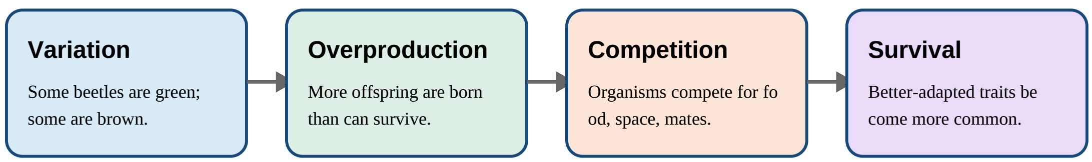
Chapter 4: Evolution and Natural Selection

Core idea: Evolution is change in populations over generations, often driven by natural selection.

Learning Goals

- Define variation, adaptation, fitness, and natural selection.
- Explain the V-O-C-S sequence.
- Interpret examples of selection in changing environments.
- Avoid the misconception that individuals evolve during their lifetime.

Natural Selection: V-O-C-S



Important: Individuals do not evolve during life. Populations evolve across generations.

Step-by-Step Explanation

1. Variation

Individuals in a population are not identical. Some differences are caused by genes, mutations, and sexual reproduction. Variation gives natural selection something to act on.

2. Overproduction and Competition

Many organisms produce more offspring than can survive. Because resources are limited, organisms compete for food, space, mates, light, or shelter.

3. Selection Over Generations

Individuals with helpful inherited traits are more likely to survive and reproduce. Over many generations, those traits become more common in the population. The population evolves; a single individual does not decide to evolve.

Key Vocabulary

Term	Meaning
Evolution	Change in inherited traits of a population over generations.
Variation	Differences among individuals in a population.
Adaptation	Inherited trait that helps survival or reproduction.

Fitness	Ability to survive and reproduce in a particular environment.
Natural selection	Process where better-adapted organisms leave more offspring.

Visual Summary and Memory Tricks

V-O-C-S: Variation, Overproduction, Competition, Survival/reproduction.

Adaptation is about fit with the environment, not being perfect. A trait that helps in one environment may hurt in another.

Virtual Lab / Investigation Practice

Virtual lab scenario: In a dark forest, dark moths are harder for birds to see. Over generations, dark moths may become more common if color is inherited.

Experimental design: To test selection, compare survival rates of different trait forms in the same environment while controlling other variables.

Exam-Style Question and Answer Session

Q1. What is natural selection?

Answer: Natural selection is the process in which organisms with helpful inherited traits survive and reproduce more successfully.

Q2. Why is variation important?

Answer: Without variation, there are no trait differences for selection to act on.

Q3. Do individuals evolve?

Answer: No. Populations evolve across generations.

Q4. What is an adaptation?

Answer: An inherited trait that improves survival or reproduction in a specific environment.

Q5. Who is strongly associated with the theory of evolution by natural selection?

Answer: Charles Darwin is strongly associated with this theory.

Harder Challenge Question

Question: A mutation makes some rabbits slightly faster. In an area with many predators, what may happen over many generations?

Answer: Faster rabbits may survive and reproduce more often, so the faster trait may become more common if it is inherited.

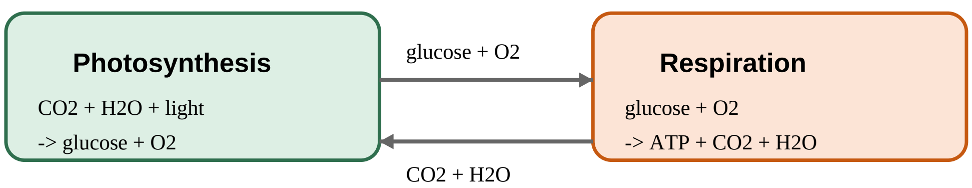
Chapter 5: Energy in Organisms - Photosynthesis and Respiration

Core idea: Life depends on energy transformations, especially photosynthesis and cellular respiration.

Learning Goals

- Write the basic word equations for photosynthesis and respiration.
- Compare glucose and ATP.
- Explain why plants and animals both use respiration.
- Connect energy flow to ecosystems.

Photosynthesis and Cellular Respiration Work Together



Memory: Photosynthesis builds sugar. Respiration breaks sugar to release ATP.

Do not confuse food energy (glucose) with immediate cell energy (ATP).

Step-by-Step Explanation

1. Photosynthesis

Photosynthesis uses carbon dioxide, water, and light energy to produce glucose and oxygen. It occurs mainly in chloroplasts. Plants use glucose for growth, storage, and energy.

2. Cellular Respiration

Cellular respiration breaks down glucose in the presence of oxygen to produce ATP, carbon dioxide, and water. It occurs in plants and animals. Plants make glucose, but they still need respiration to turn glucose into usable ATP.

3. Connecting the Processes

Photosynthesis stores energy in glucose. Respiration releases energy from glucose as ATP. The products of one process are reactants for the other, which is why these processes are often studied together.

Key Vocabulary

Term	Meaning
Photosynthesis	Process where plants, algae, and some bacteria use light to make glucose.

Cellular respiration	Process where cells break glucose to release ATP.
Glucose	Sugar molecule that stores chemical energy.
ATP	Immediate energy molecule cells use for work.
Chloroplast	Organelle where photosynthesis occurs in plants.
Mitochondrion	Organelle where much cellular respiration occurs.

Visual Summary and Memory Tricks

Photosynthesis builds food. Respiration breaks food.

Glucose is like a packed lunch; ATP is like cash the cell can spend immediately.

Plants do both photosynthesis and respiration. Animals do respiration but not photosynthesis.

Virtual Lab / Investigation Practice

Virtual lab scenario: You place one plant in bright light and another in darkness. The bright-light plant produces more oxygen bubbles because photosynthesis increases with light availability.

Data analysis: If oxygen production rises as light increases but then levels off, another factor such as carbon dioxide or temperature may become limiting.

Exam-Style Question and Answer Session

Q1. What are the reactants of photosynthesis?

Answer: Carbon dioxide, water, and light energy.

Q2. What are the products of photosynthesis?

Answer: Glucose and oxygen.

Q3. What is the purpose of cellular respiration?

Answer: To release energy from glucose and make ATP.

Q4. Do plants perform cellular respiration?

Answer: Yes. Plants need ATP just like animal cells do.

Q5. Where does photosynthesis occur in plant cells?

Answer: In chloroplasts.

Harder Challenge Question

Question: A student says, plants use photosynthesis instead of respiration. Correct the statement.

Answer: Plants use photosynthesis to make glucose, but they also use cellular respiration to break glucose and make ATP.

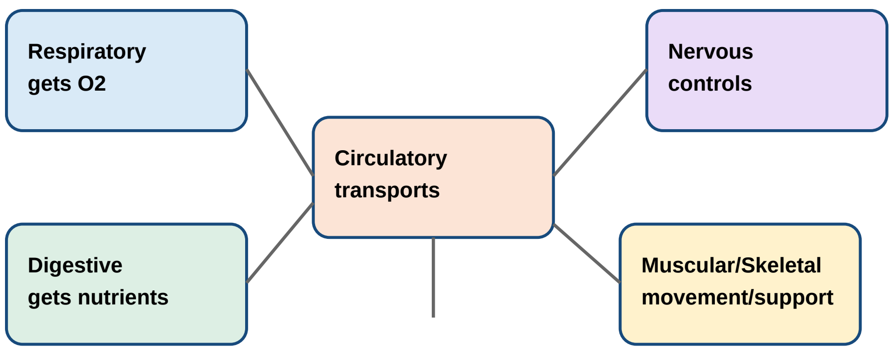
Chapter 6: Human Body Systems and Homeostasis

Core idea: Body systems work together to keep internal conditions stable and support life.

Learning Goals

- Describe major body systems and functions.
- Explain how systems interact.
- Define homeostasis with examples.
- Analyze a body-system scenario using evidence.

Human Body Systems - Teamwork Model



Bridge idea: No system works alone. Blood connects many systems by carrying oxygen, nutrients, wastes, and signals.

Step-by-Step Explanation

1. System Functions

The digestive system breaks food into molecules the body can absorb. The respiratory system exchanges gases. The circulatory system transports materials. The nervous and endocrine systems coordinate communication. The skeletal and muscular systems support movement.

2. System Interactions

After you eat, the digestive system absorbs glucose and nutrients. The circulatory system carries those nutrients to cells. The respiratory system supplies oxygen so cells can perform respiration. The excretory system removes wastes from metabolism.

3. Homeostasis

Homeostasis means keeping internal conditions within a healthy range. Examples include body temperature regulation, blood glucose control, and water balance. When you exercise, your breathing and heart rate increase to deliver more oxygen and remove carbon dioxide.

Key Vocabulary

Term	Meaning

Homeostasis	Maintenance of stable internal conditions.
Digestive system	Breaks food into nutrients.
Respiratory system	Brings in oxygen and removes carbon dioxide.
Circulatory system	Transports blood, oxygen, nutrients, wastes, and signals.
Nervous system	Senses information and coordinates responses.
Excretory system	Removes wastes and helps balance water and salts.

Visual Summary and Memory Tricks

D-C-R-N: Digestive Collects nutrients, Circulatory Routes materials, Respiratory gets oxygen, Nervous Notices and controls.

Homeostasis = balance. Think of a thermostat keeping a room near the target temperature.

Virtual Lab / Investigation Practice

Virtual lab scenario: During exercise, heart rate and breathing rate rise. This is not random; it helps deliver more oxygen and glucose to muscle cells and remove carbon dioxide.

Case analysis: If someone has low oxygen levels, both the respiratory and circulatory systems are involved because oxygen must enter the lungs and then travel in blood.

Exam-Style Question and Answer Session

Q1. What does the circulatory system do?

Answer: It transports blood, oxygen, nutrients, wastes, and chemical signals.

Q2. What does the digestive system do?

Answer: It breaks food into nutrients that can be absorbed.

Q3. What is homeostasis?

Answer: The maintenance of stable internal conditions.

Q4. Why does breathing rate increase during exercise?

Answer: Muscle cells need more oxygen for respiration and need to remove more carbon dioxide.

Q5. Which system controls quick responses to stimuli?

Answer: The nervous system.

Harder Challenge Question

Question: Explain how at least three systems work together after eating a meal.

Answer: The digestive system breaks food into nutrients, the circulatory system transports nutrients to cells, and the respiratory system supplies oxygen so cells can release energy from those nutrients.

Supplementary Study Material

Common Misconceptions to Fix Before High School Biology

Misconception	Correct Understanding
Plants do not need respiration.	Plants do cellular respiration because all cells need ATP.
Evolution means an individual changes because it wants to.	Populations evolve across generations due to inherited variation and selection.
All bacteria are harmful.	Many bacteria are helpful, including decomposers and gut bacteria.
Dominant traits are always more common.	Dominant means expressed with one allele, not necessarily common.
Food chains show all ecosystem relationships.	Food chains are simplified; food webs are more realistic.

One-Page Memory Sheet

Topic	Memory Trick	Must-Know Idea
Cells	Cell = City	Organelles have jobs that support cell life.
DNA	A-T Apple Tree; C-G Car Garage	DNA instructions influence proteins and traits.
Ecosystems	Sun starts everything	Energy flows; nutrients recycle.
Evolution	V-O-C-S	Variation plus selection changes populations.
Energy	Build vs break	Photosynthesis builds glucose; respiration makes ATP.
Body Systems	Homeostasis = balance	Systems interact to keep the body stable.

Mini Glossary

Term	Definition
Adaptation	Inherited trait that helps survival/reproduction.
Allele	Different form of a gene.
ATP	Immediate energy molecule for cells.
Consumer	Organism that gets energy by eating other organisms.
Decomposer	Organism that breaks down dead matter.
Gene	DNA segment that influences a trait.
Homeostasis	Stable internal balance.
Mutation	Change in DNA sequence.
Producer	Organism that makes its own food.
Trait	Characteristic of an organism.

Four-Week Bridge Course Study Plan

Week	Focus	Tasks
1	Cells and cell processes	Read Chapter 1. Redraw cell diagram. Answer questions without notes.

2	DNA and heredity	Read Chapter 2. Practice base pairing and Punnett squares.
3	Ecosystems and evolution	Read Chapters 3-4. Explain food webs and V-O-C-S aloud.
4	Energy and body systems	Read Chapters 5-6. Take the mock exam and review mistakes.

Full Mock Exam - High School Biology Bridge Course

Instructions: Try this like a real exam. Do not look at the answer key until the end. Write complete sentences for short-answer questions. For application questions, explain your reasoning, not just the final answer.

Section A: Multiple Choice

1. Which organelle stores DNA in a eukaryotic cell?
A. Ribosome B. Nucleus C. Mitochondrion D. Cytoplasm

2. A bacterial cell is usually classified as what type?
A. Eukaryotic B. Prokaryotic C. Multicellular D. Plant

3. Which base pairs with guanine in DNA?
A. Adenine B. Thymine C. Cytosine D. Uracil

4. What is a gene?
A. Whole organism B. Segment of DNA C. Cell membrane D. Ecosystem

5. Which is abiotic?
A. Grass B. Mushroom C. Water D. Bacteria

6. Which organism is usually a producer?
A. Plant B. Fox C. Rabbit D. Fungus

7. Natural selection acts most directly on what?
A. Inherited variation B. Wishes C. Learned habits only D. Nonliving rocks

8. What is produced during photosynthesis?
A. Glucose and oxygen B. Carbon dioxide and water C. ATP only D. Nitrogen gas

9. What is the main purpose of cellular respiration?
A. To store sunlight B. To make ATP C. To pair DNA bases D. To make soil

10. What is homeostasis?
A. Cell division B. Internal balance C. DNA mutation D. Food web

Section B: Short Answer

11. Explain two differences between prokaryotic and eukaryotic cells.

12. A DNA strand is A T G C C A. Write the complementary strand.

13. Explain why decomposers are important in ecosystems.

14. Describe the V-O-C-S model of natural selection.

15. Explain why plants need both photosynthesis and respiration.

16. Give one example of two human body systems working together.

Section C: Data and Scenario Questions

17. A pond receives fertilizer runoff. Algae grow rapidly, then fish begin to die. Explain the likely chain of events.

18. In a population of insects, some are resistant to a pesticide. After many pesticide treatments, most surviving insects are resistant. Explain this using natural selection.

19. A student observes a cell with a nucleus, mitochondria, chloroplasts, and a cell wall. Identify the cell type and justify your answer.

20. During a race, a runner breathes faster and the heart beats faster. Explain why using cellular respiration and body systems.

Answer Key and Explanations

1. B. Which organelle stores DNA in a eukaryotic cell?
2. B. A bacterial cell is usually classified as what type?
3. C. Which base pairs with guanine in DNA?
4. B. What is a gene?
5. C. Which is abiotic?
6. A. Which organism is usually a producer?
7. A. Natural selection acts most directly on what?
8. A. What is produced during photosynthesis?
9. B. What is the main purpose of cellular respiration?
10. B. What is homeostasis?
11. Prokaryotes lack a nucleus and are usually simpler; eukaryotes have a nucleus and membrane-bound organelles.
12. T A C G G T.
13. They break down dead matter and recycle nutrients back into ecosystems.
14. Variation, overproduction, competition, and survival/reproduction of better-adapted organisms.
15. Photosynthesis makes glucose; respiration breaks glucose to make ATP.
16. Example: respiratory brings oxygen in and circulatory transports it to cells.
17. Fertilizer adds nutrients, causing algal bloom; decomposers use oxygen breaking down dead algae; low oxygen can kill fish.
18. Resistant insects survive and reproduce, increasing resistance alleles in the population.
19. It is a plant eukaryotic cell because it has a nucleus plus chloroplasts and a cell wall.
20. Muscles need more ATP, so cells need more oxygen and glucose; respiratory and circulatory systems increase delivery and waste removal.

Harder Biology Challenge Questions

These are AMC-style in the sense that they test reasoning, patterns, and careful interpretation rather than simple memorization. Biology competitions usually call these application or analysis questions.

Challenge 1. A cell produces large amounts of protein for export. Which organelles would you expect to be especially active?

Answer: Ribosomes make proteins, rough ER helps process/transport them, Golgi apparatus modifies/packages them, and mitochondria provide ATP.

Challenge 2. Two parents are Aa and aa. What fraction of children are expected to show the recessive phenotype if A is dominant?

Answer: Half. Cross Aa x aa gives Aa and aa in a 1:1 ratio.

Challenge 3. A food web has grass, grasshoppers, frogs, snakes, and hawks. If frogs decrease sharply, predict two effects.

Answer: Grasshoppers may increase because fewer frogs eat them. Snakes or hawks may decrease if they rely on frogs or snakes as food.

Challenge 4. Why can a helpful mutation become harmful after an environment changes?

Answer: Fitness depends on environment. A trait helpful in one environment may reduce survival in another.

Challenge 5. A plant grows well in light but poorly in darkness even with water and minerals. What missing factor explains this?

Answer: Light is needed for photosynthesis, so the plant cannot make enough glucose in darkness.

Challenge 6. A person has a respiratory illness that reduces oxygen exchange. Why might muscles feel weak?

Answer: Less oxygen reaches cells, reducing aerobic respiration and ATP production.

Challenge 7. A graph shows predator population rising shortly after prey population rises. Explain the lag.

Answer: More prey provides more food, allowing predator survival/reproduction to increase after some time.

Challenge 8. A DNA mutation changes one base but the organism looks the same. Give two reasons.

Answer: The mutation may not change the protein, or the changed protein may still function normally; the trait may also be influenced by other genes/environment.

Challenge 9. Why is biodiversity often linked with ecosystem stability?

Answer: More species can provide backup roles and alternative food sources, making the system less likely to collapse after one change.

Challenge 10. Explain why antibiotics can lead to resistant bacteria becoming common.

Answer: Susceptible bacteria die, resistant bacteria survive and reproduce, so resistance becomes more common in the population.

Flashcards for Fast Review

Front	Back
What does the nucleus do?	Stores DNA and controls many cell activities.
What does mitochondrion do?	Produces ATP through cellular respiration.
A pairs with?	T.
C pairs with?	G.
Producer example	Plant, algae, or photosynthetic bacteria.
Decomposer example	Fungus or bacteria.
V-O-C-S	Variation, Overproduction, Competition, Survival/reproduction.
Photosynthesis word equation	Carbon dioxide + water + light -> glucose + oxygen.
Respiration word equation	Glucose + oxygen -> ATP + carbon dioxide + water.
Homeostasis	Stable internal balance.

Final Exam Readiness Checklist

- I can draw and label an animal cell and a plant cell.
- I can explain DNA base pairing and simple inheritance.
- I can predict what happens when one part of a food web changes.
- I can explain natural selection without saying organisms evolve because they want to.
- I can compare photosynthesis and respiration.
- I can explain how body systems work together to maintain homeostasis.
- I can answer scenario questions using evidence and reasoning.

End of Textbook

Tip: Review your wrong answers first. Mistakes show exactly where your next study session should focus.